Si5351A VFO Rev A — Field Manual Ch. 6. Plan: PLAN.md (2026-10-22). Hierarchy: power/USB, MCU/UI, synth/RF.
Inter-sheet nets via global labels (MCU_TX, MCU_RX, DTR, I2C_SCL, I2C_SDA) and power rails.

Power and USB

File: power_usb.kicad_sch

MCU and UI

File: mcu_ui.kicad_sch

Synth and RF outputs

File: synth_rf.kicad_sch

Draft schematic, not reviewed. See VERIFY notes.

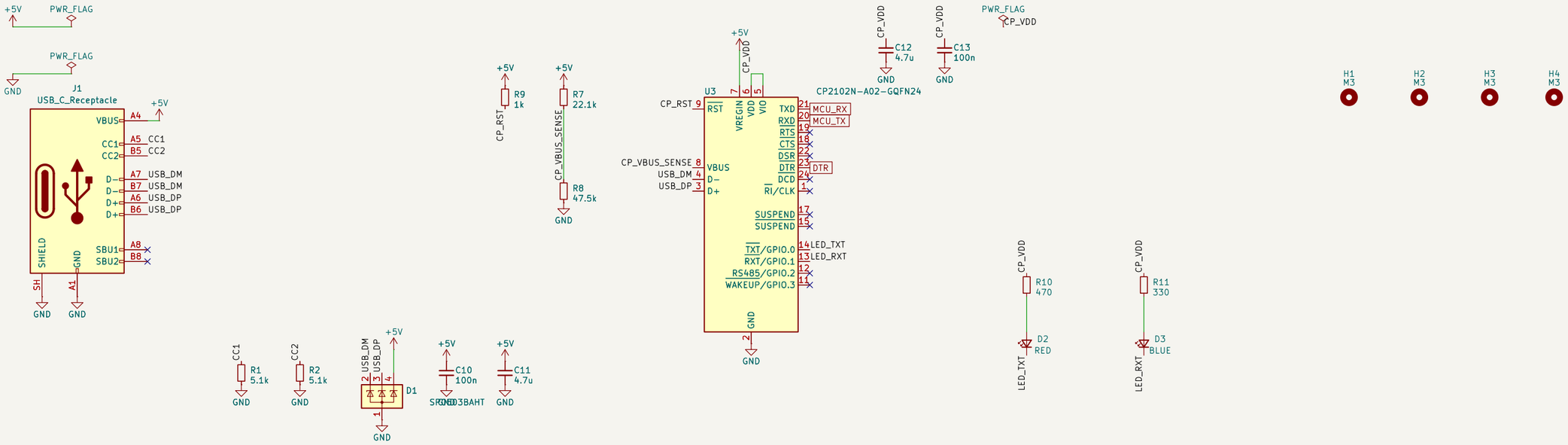
Etherkit / NT7S

Sheet: /
File: Si5351AVFO.kicad_sch

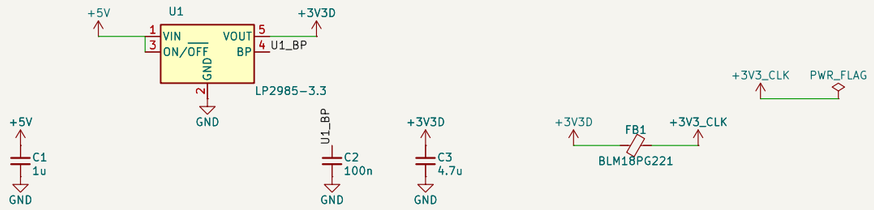
Title: Si5351A VFO — Top

Size: A3	Date: 2026-10-22	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/4

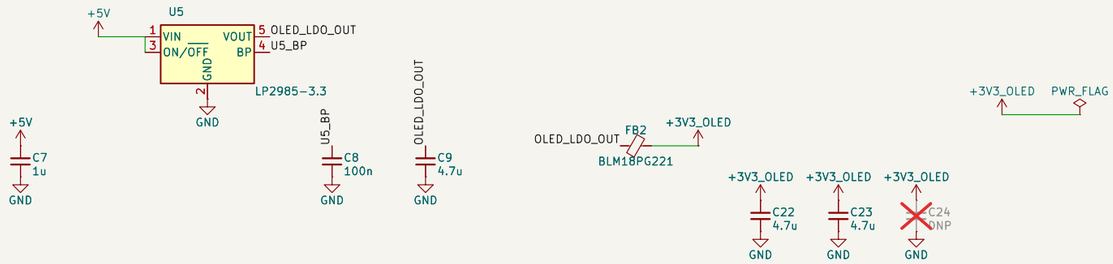
USB-C input, CP2102N USB-UART (from EtherkitUARTAdapter), LP2985 rails (U1 from Rev F; U5 dedicated to OLED per Jason 2026-10-22)



U1: +3V3D (MCU) and, via FB1, +3V3_CLK (Si5351 + TCXO). Rev F U1/C1/C2/C3.



U5: OLED-only rail (Eyes On The Prize lesson, PLAN 4a). Place U5/FB2/C22-C24 at the OLED header.



VERIFY (unconfirmed items on this sheet):
- J1: UART adapter schematic pairs GCT USB4105 footprint with this HCTL LCSC part; HCTL footprint used here — confirm. Shield to GND assumed (not stated in PLAN).
- D1: Pin-to-line mapping (D-/D+/VBUS on pins 2/3/4) to be checked against adapter layout.
- C10: LCSC stock low; fallback C49678 (0805) per PLAN
- U3: Unused CP2102N pins left NC (adapter used DTR via JP1, TRRS dropped). Confirm none were strapped on the adapter.
- R9: PLAN says /RST pull-up to VBUS (as on adapter); CP2102N RST max is VIO-referenced — confirm adapter really ties to VBUS.
- D2: LED drive topology (VDD->R->LED->GPIO active-low) assumed; PLAN only says CP2102N pin 14. Optional per PLAN Q5.
- D3: Same topology assumption as D2 (pin 13). Optional per PLAN Q5.
- FB2: Bead or series R (<=6 ohm upper bound, PLAN 4a); value to be set by measurement.
- C22: OLED bulk value: PLAN start point 2x4.7u, tune by measurement
- C24: DNP 1206 site for larger OLED bulk cap; value/part not chosen

Draft schematic, not reviewed. See VERIFY notes.

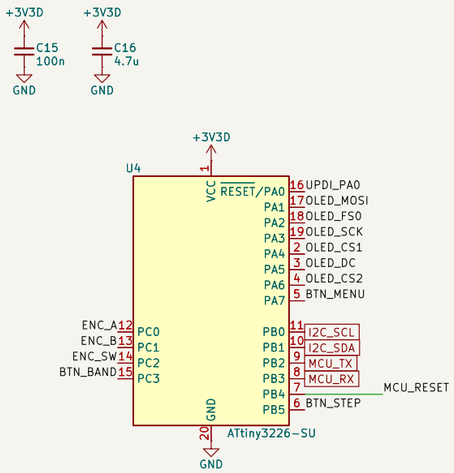
Etherkit / NT7S

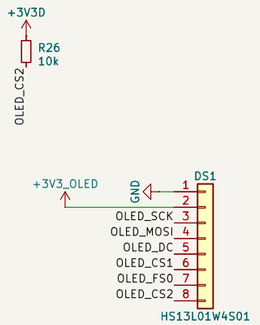
Sheet: /Power and USB/
File: power_usb.kicad_sch

Title: Si5351A VFO — Power and USB

Size: A3	Date: 2026-10-22	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 2/4

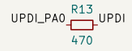
ATtiny3226 @ 8 MHz internal, 3.3 V. Optiboot over CP2102N; DTR auto-reset on PB4 (RSTPINCFG=0x3, megaTinyCore "UPDI w/reset on PB4"), as EtherKeyerMini C3/R4.

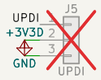


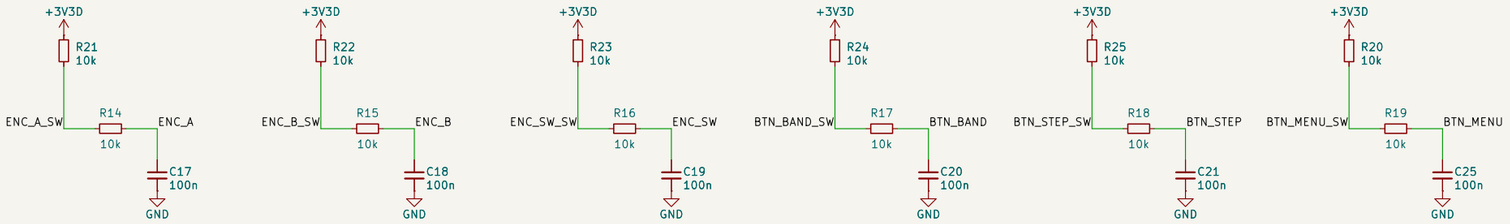


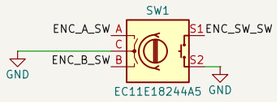
OLED: SPI only (PA1 MOSI, PA3 SCK, PA4 CS1, PA5 DC). Never shares the Si5351 I2C bus.
CS2 (font chip) held high by R26.

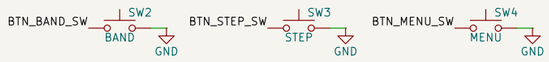












VERIFY (unconfirmed items on this sheet):
- U4: Pin mux (PLAN sec.3) to be checked vs datasheet I/O mux table. Jason consigning parts.
- C14: PLAN: series R or clamp on PB4 for DTR-rising overshoot? EKM has none.
- J5: Generic header, no part number (PLAN). Pin order UPDI/GND/3V3 per PLAN.
- DS1: 1.3in SPI SH1106 module. Pin order GND,VCC,SCL,SDA,DC,CS1,FS0,CS2 from PLAN pin list; pin numbering, pitch and connector style unverified. Low stock - Jason sourcing.
- C17: Debounce topology/count: PLAN lists x5 but 6 inputs (A,B,SW,3 buttons) - drawn as 6. Check vs encoder bounce spec.
- SW1: Detent/pulse count unverified.

Draft schematic, not reviewed. See VERIFY notes.

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Sheet: /MCU and UI/
File: mcu_ui.kicad_sch

Title: Si5351A VFO - MCU and UI

Size: A3	Date: 2026-10-22	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 3/4

Si5351A + 25 MHz TCXO copied from Si5351A Breakout Rev F (R5/C5/C6/R6/C4, XB open). Each output: Rs 0R -> Cb 100n DC block -> DNP shunt -> LPF (DNP) with 0R bypass -> SMA.

VERIFY (unconfirmed items on this sheet):
- X1: TCXO stock thin (718): buy before ordering.
- R27: I2C pull-up value not fixed (PLAN 2.2-4.7k); no LCSC chosen.
- R28: as R27
- R100: 0603 0R series (Jason decision): LCSC code not in PLAN (C17477 is 0805).
- C100: DC-block LF corner ~32 kHz into 50 ohm (PLAN Q6).
- R101: DNP shunt ~120 ohm back-match of ~85 ohm source (datasheet Zo); part/LCSC not chosen; verify return loss esp. >100 MHz.
- C101: LPF values builder choice (DNP)
- L1: LPF inductor size (SMD vs T37 option, PLAN Q7)
- C105: Parallel C across L (elliptic) per LowPassFilterModule - unconfirmed
- J2: Footprint is a placeholder; check BWSMA-KE-P001 launch geometry and 1.6 mm edge fit.
- R120: see R100
- R121: see R100
- J3: see J2
- R140: see R100
- R141: see R100
- J4: see J2

+3V3_CLK
R5
10
TCXO_VDD
PWR_FLAG
TCXO_VDD
C5
100n
GND
+3V3_CLK
C4
100n
GND
TCXO_EN
R6
0
X1
25MHz TCXO
TCXO_VDD
TCXO_EN 1
NC/EN
VDD 4
GND 2
OUT 3
TCXO_OUT
GND
TCXO_OUT
C6
100n
SI_XA
+3V3_CLK
U2
SI_XA 2
XA
XB 3
VDD 1
VDDO 7
CLK0 10
CLK0
CLK1 9
CLK1
CLK2 6
CLK2
I2C_SCL 4
SCL
I2C_SDA 5
SDA
GND 8
GND
Si5351A-B-GT
+3V3_CLK
R27
VERIFY 2.2k-4.7k
I2C_SCL
+3V3_CLK
R28
VERIFY 2.2k-4.7k
I2C_SDA

Bypass: RC (R104/R124/R144) fitted by default; RA/RB DNP. Filter build: fit RA/RB, remove RC.
Keep RA/RB pads <=2 mm from junction (PLAN 6).

CLK0 output chain (J2)

CLK0 R100 0 CLK0_S
CLK0_S C100 100n CLK0_DC
CLK0_DC R101 120 GND
CLK0_DC R102 0 CLK0_F1
CLK0_F1 C101 DNP GND
CLK0_F1 C105 DNP CLK0_F2
CLK0_F1 L1 DNP CLK0_F2
CLK0_F2 C102 DNP GND
CLK0_F2 C106 DNP CLK0_F3
CLK0_F2 L2 DNP CLK0_F3
CLK0_F3 C103 DNP GND
CLK0_F3 C107 DNP CLK0_F4
CLK0_F3 L3 DNP CLK0_F4
CLK0_F4 C104 DNP GND
CLK0_F4 R103 0 CLK0_OUT
CLK0_DC R104 0 CLK0_OUT
CLK0_OUT J2 SMA CLK0 GND

CLK1 output chain (J3)

CLK1 R120 0 CLK1_S
CLK1_S C120 100n CLK1_DC
CLK1_DC R121 120 GND
CLK1_DC R122 0 CLK1_F1
CLK1_F1 C121 DNP GND
CLK1_F1 C125 DNP CLK1_F2
CLK1_F1 L4 DNP CLK1_F2
CLK1_F2 C122 DNP GND
CLK1_F2 C126 DNP CLK1_F3
CLK1_F2 L5 DNP CLK1_F3
CLK1_F3 C123 DNP GND
CLK1_F3 C127 DNP CLK1_F4
CLK1_F3 L6 DNP CLK1_F4
CLK1_F4 C124 DNP GND
CLK1_F4 R123 0 CLK1_OUT
CLK1_DC R124 0 CLK1_OUT
CLK1_OUT J3 SMA CLK1 GND

CLK2 output chain (J4)

CLK2 R140 0 CLK2_S
CLK2_S C140 100n CLK2_DC
CLK2_DC R141 120 GND
CLK2_DC R142 0 CLK2_F1
CLK2_F1 C141 DNP GND
CLK2_F1 C145 DNP CLK2_F2
CLK2_F1 L7 DNP CLK2_F2
CLK2_F2 C142 DNP GND
CLK2_F2 C146 DNP CLK2_F3
CLK2_F2 L8 DNP CLK2_F3
CLK2_F3 C143 DNP GND
CLK2_F3 C147 DNP CLK2_F4
CLK2_F3 L9 DNP CLK2_F4
CLK2_F4 C144 DNP GND
CLK2_F4 R143 0 CLK2_OUT
CLK2_DC R144 0 CLK2_OUT
CLK2_OUT J4 SMA CLK2 GND

Draft schematic, not reviewed. See VERIFY notes.
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Sheet: /Synth and RF outputs/
File: synth_rf.kicad_sch

Title: Si5351A VFO - Synth and RF outputs

Size: A3	Date: 2026-10-22	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 4/4